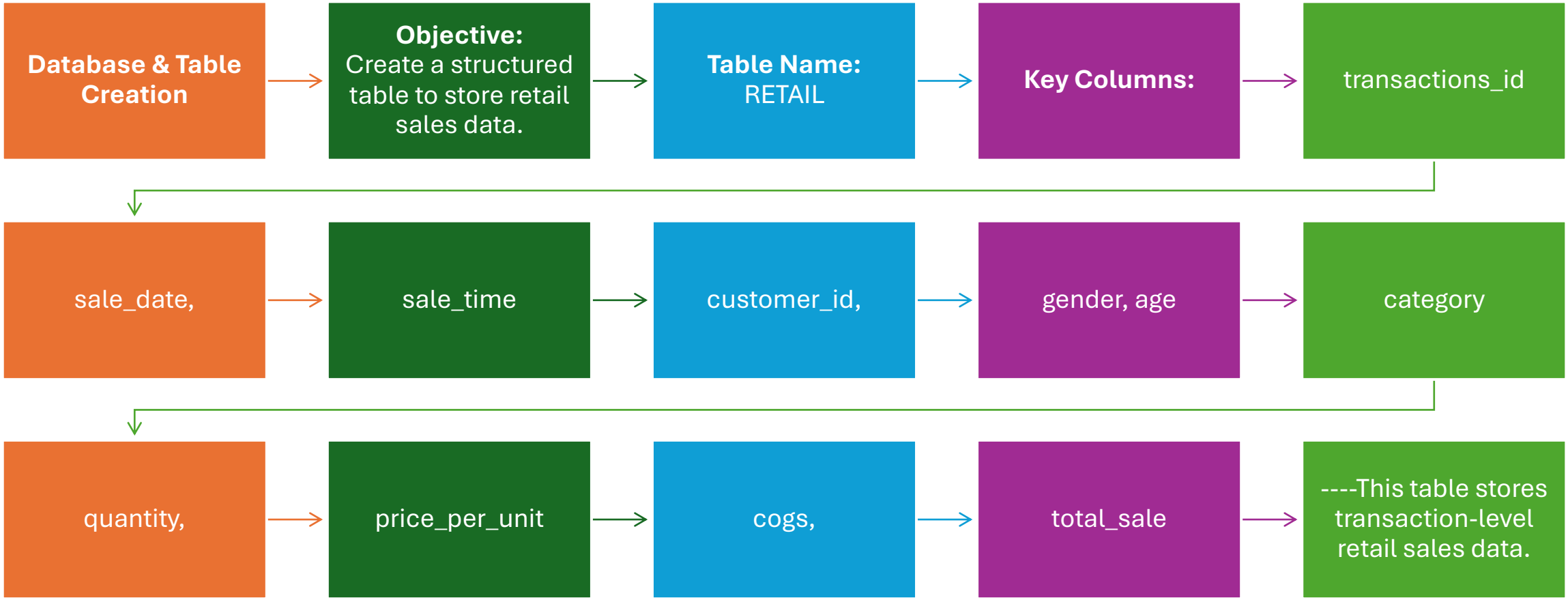
Retail Sales Data Analysis using SQL

Project Overview

This project focuses on analyzing retail sales data using SQL to extract meaningful business insights. The dataset contains transactional-level information such as sales date, time, customer details, product categories, quantity sold, and total sales. The goal of this project was to clean raw data, perform exploratory analysis, and solve real-world business problems using SQL queries.


Database & Table Creation
Objective: Create a structured table to store retail sales data.
Table Name: RETAIL
Key Columns:
transactions_id
sale_date,
sale_time
customer_id,
gender, age
category
quantity,
price_per_unit
cogs,
total_sale
----This table stores transaction-level retail sales data.

Data Cleaning – Removing NULL Records

Objective:
Remove inaccurate or incomplete records.

Action Taken:

- Deleted rows containing NULL values in important columns.
- Ensured clean and consistent data for analysis.
- This step improves accuracy of insights.

```
--DELETE NULL VALUES
delete from RETAIL
WHERE transactions_id is null
or
 sale_date is null
or
customer_id is null
or
gender is null
or
category is null
or
 quantity is null
or
price_per_unit  is null
or
 cogs is null
or
 total_sale is null;
```

```
select * from RETAIL
WHERE transactions_id is null
or
 sale_date is null
or
customer_id is null
or
gender is null
or
category is null
or
 quantity is null
or
price_per_unit  is null
or
 cogs is null
```

Data Insights – Basic Analysis

Key Questions Answered:

- Total number of sales transactions
- Total number of unique customers
- Total number of unique product categories

Purpose:

- Understand dataset size and structure
- Get an overview of sales activity

```
                --DATA INSIGHT --

-- Q1. HOW MANY SALES WE HAVE
SELECT count(*) as total_sale FROM RETAIL;

-- Q2.HOW MANY CUSTOMERS WE HAVE
SELECT count(DISTINCT customer_id) as customer_id from RETAIL;

--Q3. HOW  MANY UNIQUE CATEGORY WE HAVE
SELECT  DISTINCT category as category from RETAIL;
```

Query 1 – Sales on a Specific Date

Problem Statement:
Retrieve all sales made on **05-Nov-2022**.

Insight Gained:

- Helped analyze daily sales performance
- Useful for event-based or festive sales analysis

```
68
69  --Q.1 Write a SQL query to retrieve all columns for sales made on '2022-11-05
70
71  select * from RETAIL
72  where sale_date = TO_DATE ('2022-11-05' ,'yyyy-mm-dd');
```

Data Output | Messages | Notifications

Showing rows: 1 to 11 | Page No: 1 of 1

	transactions_id integer	sale_date date	sale_time time without time zone	customer_id integer	gender character varying (20)	age integer	category character varying (15)	quantity integer	price_per_unit double precision	cogs double precision	total_sale double precision
1	180	2022-11-05	10:47:00	117	Male	41	Clothing	3	300	129	900
2	240	2022-11-05	11:49:00	95	Female	23	Beauty	1	300	123	300
3	1256	2022-11-05	09:58:00	29	Male	23	Clothing	2	500	190	1000
4	1587	2022-11-05	20:06:00	140	Female	40	Beauty	4	300	105	1200
5	1819	2022-11-05	20:44:00	83	Female	35	Beauty	2	50	13.5	100
6	943	2022-11-05	19:29:00	90	Female	57	Clothing	4	300	318	1200
7	1896	2022-11-05	20:19:00	87	Female	30	Electronics	2	25	30.75	50
8	1137	2022-11-05	22:34:00	104	Male	46	Beauty	2	500	145	1000
9	856	2022-11-05	17:43:00	102	Male	54	Electronics	4	30	9.3	120
10	214	2022-11-05	16:31:00	53	Male	20	Beauty	2	30	8.1	60
11	1265	2022-11-05	14:35:00	86	Male	55	Clothing	3	300	111	900

Query 3 – Category-wise Sales Analysis

Problem Statement:

Calculate:

- Total sales for each category
- Total number of orders per category

Insights:

- Identifies high-performing product categories
- Helps in inventory and marketing decisions

```
81  --Q.3 Write a SQL query to calculate the total sales (total_sale) for each category.
82  select category ,
83  sum(total_sale) as total ,
84  count(*) as total_order
85  from RETAIL
86  group by 1 ;
87
```

Data Output | Messages | Notifications

SQL

	category character varying (15)	total double precision	total_order bigint
1	Electronics	313810	684
2	Clothing	311070	701
3	Beauty	286840	612

Query 4 – High-Value Transactions

Problem Statement:
Find transactions where total sale > 1000.

Business Impact:

- Identifies premium customers
- Useful for loyalty programs and targeted offers

```
88  -- Q.5 Write a SQL query to find all transactions where the total_sale is greater than 1000.
89  select * from RETAIL
90  where total_sale >1000;
```

Data Output | Messages | Notifications

Showing rows: 1 to 306 | Page No: 1 of 1

	transactions_id integer	sale_date date	sale_time time without time zone	customer_id integer	gender character varying (20)	age integer	category character varying (15)	quantity integer	price_per_unit double precision	cogs double precision	total_sale double precision
1	522	2022-07-09	11:00:00	52	Male	46	Beauty	3	500	145	1500
2	559	2022-12-12	10:48:00	5	Female	40	Clothing	4	300	84	1200
3	1522	2022-11-14	08:35:00	48	Male	46	Beauty	3	500	235	1500
4	1559	2022-08-20	07:40:00	49	Female	40	Clothing	4	300	144	1200
5	421	2022-04-08	08:43:00	66	Female	37	Clothing	3	500	235	1500
6	1421	2022-01-17	07:07:00	59	Female	37	Clothing	3	500	185	1500
7	484	2022-03-13	07:52:00	135	Female	19	Clothing	4	300	75	1200
8	1484	2022-11-23	09:29:00	22	Female	19	Clothing	4	300	147	1200
9	15	2022-07-01	11:50:00	75	Female	42	Electronics	4	500	210	2000
10	743	2022-08-07	07:54:00	55	Female	34	Beauty	4	500	260	2000
11	1015	2022-03-09	11:53:00	94	Female	42	Electronics	4	500	200	2000
12	1743	2022-10-26	09:37:00	47	Female	34	Beauty	4	500	250	2000
13	742	2022-03-19	06:08:00	37	Female	38	Electronics	4	500	195	2000
14	1742	2022-11-22	08:25:00	18	Female	38	Electronics	4	500	220	2000
15	420	2022-01-02	10:53:00	28	Female	22	Clothing	4	500	200	2000
16	1420	2022-04-15	07:01:00	138	Female	22	Clothing	4	500	205	2000
17	592	2022-12-26	09:15:00	77	Female	46	Beauty	4	500	275	2000
18	1592	2022-03-16	09:08:00	81	Female	46	Beauty	4	500	155	2000
19	720	2022-04-08	08:50:00	116	Female	56	Beauty	3	500	235	1500
20	1720	2022-10-10	08:51:00	28	Female	56	Beauty	3	500	190	1500

Total rows: 306 | Query complete 00:00:00.273 | CRLF | Ln 97, Col 12

Query 5 – Gender-wise Transactions

Problem Statement:
Find total transactions made by:

- Each gender
- In each product category

Insights:

- Understand customer demographics
- Helps in gender-focused marketing strategies

```
92  -- Q.6 Write a SQL query to find the total number of transactions (transaction_id) made by each gender in each category.
93  select category ,gender ,
94  count(*) transaction_id
95  from RETAIL
96  GROUP BY category,gender
97  order by 1;
98
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1

	category character varying (15)	gender character varying (20)	transaction_id bigint
1	Beauty	Female	330
2	Beauty	Male	282
3	Clothing	Female	347
4	Clothing	Male	354
5	Electronics	Male	344
6	Electronics	Female	340

Advanced Analysis – Best Selling Month per Year

Problem Statement:

- Calculate average monthly sales
- Identify best-selling month for each year

SQL Concepts Used:

- Date extraction
- Window function (RANK)

Business Value:

- Seasonal sales trend analysis
- Better planning for promotions

```sql
--Q.7 Write a SQL query to calculate the average sale for each month. Find out best selling month in each year
select year,month,avg_sale
from
(select
extract(year from sale_date) as year,
extract (month from sale_date) as month,
avg(total_sale) as avg_sale,
rank() over (partition by extract(year from sale_date) order by avg(total_sale)desc) as rank
from RETAIL
GROUP BY 1,2
)AS T1
WHERE RANK =1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2

	year numeric	month numeric	avg_sale double precision
1	2022	7	541.3414634146342
2	2023	2	535.531914893617

Advanced Analysis – Top 5 Customers

Problem Statement:
Identify top 5 customers based on highest total sales.

Insights:

- Recognizes high-value customers
- Supports customer retention strategies

```
111
112  --Q.8 Write a SQL query to find the top 5 customers based on the highest total sales
113  select customer_id,
114  sum(total_sale)as top_sale
115  from RETAIL
116  group by 1
117  order by 2 DESC
118  limit 5;
119
```

Data Output | Messages | Notifications

SQL

	customer_id integer	top_sale double precision
1	3	38440
2	1	30750
3	5	30405
4	2	25295
5	4	23580

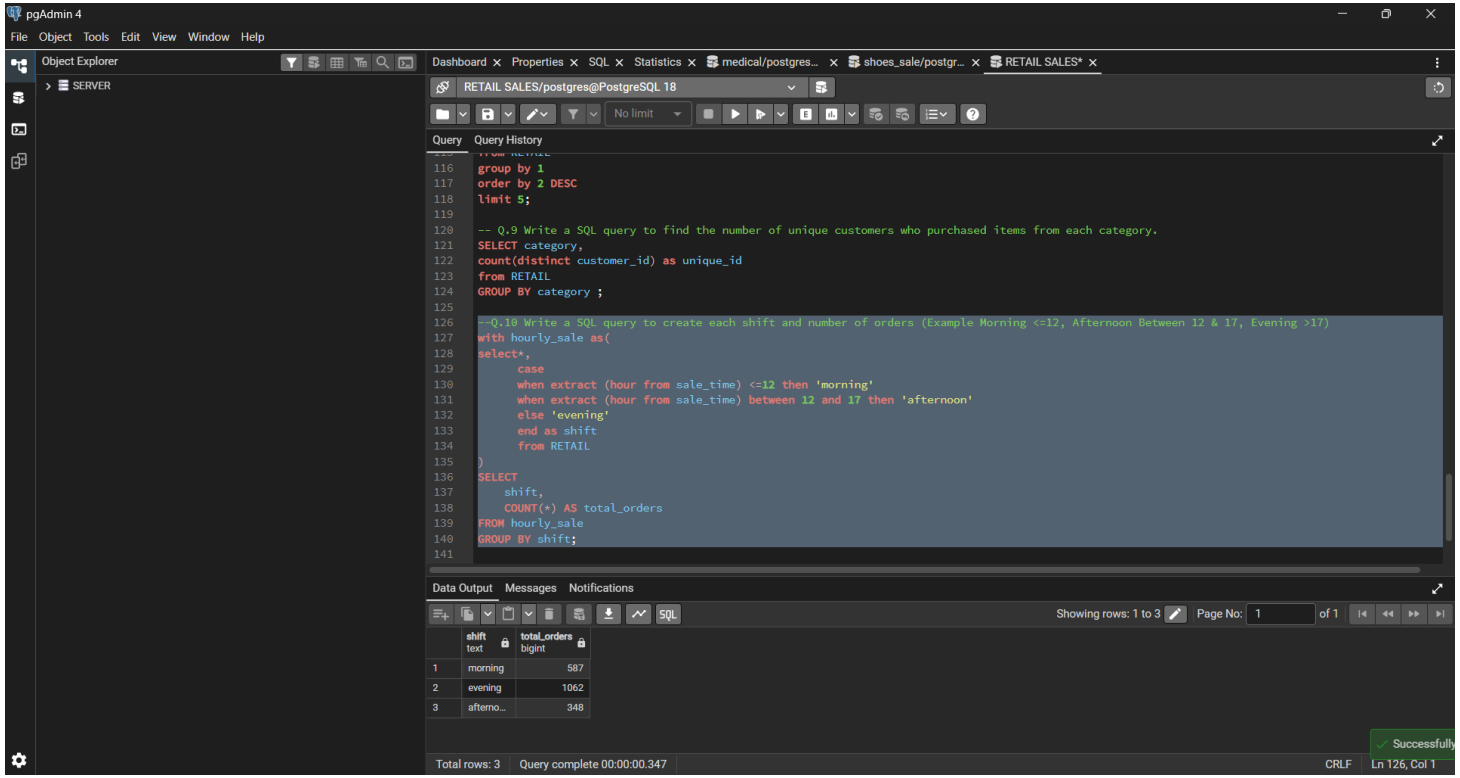
Time-Based Analysis – Shift-wise Orders

Problem Statement:
Divide sales into shifts:

- Morning (≤ 12)
- Afternoon (12–17)
- Evening (>17)

Insights:

- Identifies peak order time
- Helps in staff and resource planning



Conclusion & Learnings

Clean data is crucial for accurate analysis

SQL can solve real-world business problems

Gained hands-on experience with:

- Data cleaning
- Business analysis
- Advanced SQL queries

This project strengthens my Data Analyst portfolio.